

Model	#Param	PPL
Grave et al. (2016b) - LSTM	-	48.7
Bai et al. (2018) - TCN	-	45.2
Dauphin et al. (2016) - GCNN-8	-	44.9
Grave et al. (2016b) - LSTM + Neural cache	-	40.8
Dauphin et al. (2016) - GCNN-14	-	37.2
Merity et al. (2018) - QRNN	151M	33.0
Rae et al. (2018) - Hebbian + Cache	-	29.9
Ours - Transformer-XL Standard	151M	24.0
Baevski and Auli (2018) - Adaptive Input [◊]	247M	20.5
Ours - Transformer-XL Large	257M	18.3

Table 1: Comparison with state-of-the-art results on WikiText-103. [◊] indicates contemporary work.

Model	#Param	bpc
Ha et al. (2016) - LN HyperNetworks	27M	1.34
Chung et al. (2016) - LN HM-LSTM	35M	1.32
Zilly et al. (2016) - RHN	46M	1.27
Mujika et al. (2017) - FS-LSTM-4	47M	1.25
Krause et al. (2016) - Large mLSTM	46M	1.24
Knol (2017) - cmix v13	-	1.23
Al-Rfou et al. (2018) - 12L Transformer	44M	1.11
Ours - 12L Transformer-XL	41M	1.06
Al-Rfou et al. (2018) - 64L Transformer	235M	1.06
Ours - 18L Transformer-XL	88M	1.03
Ours - 24L Transformer-XL	277M	0.99

Table 2: Comparison with state-of-the-art results on enwik8.

modeling to have a comparison with state-of-the-art systems, including WikiText-103 (Merity et al., 2016), enwik8 (LLC, 2009), text8 (LLC, 2009), One Billion Word (Chelba et al., 2013), and Penn Treebank (Mikolov and Zweig, 2012).

WikiText-103 is the largest available word-level language modeling benchmark with long-term dependency. It contains 103M training tokens from 28K articles, with an average length of 3.6K tokens per article, which allows testing the ability of long-term dependency modeling. We set the attention length to 384 during training and 1600 during evaluation. We adopted adaptive softmax and input representations (Baevski and Auli, 2018; Grave et al., 2016a). As shown in Table 1, Transformer-XL reduces the previous state-of-the-art (SoTA) perplexity from 20.5 to 18.3, which demonstrates the superiority of the Transformer-XL architecture.

The dataset enwik8 contains 100M bytes of unprocessed Wikipedia text. We compare our architecture with the previous results in Table 2. Under the model size constraint, the 12-layer Transformer-XL achieves a new SoTA result, outperforming the 12-layer vanilla Transformer from Al-Rfou et al. (2018) by 0.05, while both Trans-

Model	#Param	bpc
Cooijmans et al. (2016) - BN-LSTM	-	1.36
Chung et al. (2016) - LN HM-LSTM	35M	1.29
Zilly et al. (2016) - RHN	45M	1.27
Krause et al. (2016) - Large mLSTM	45M	1.27
Al-Rfou et al. (2018) - 12L Transformer	44M	1.18
Al-Rfou et al. (2018) - 64L Transformer	235M	1.13
Ours - 24L Transformer-XL	277M	1.08

Table 3: Comparison with state-of-the-art results on text8.

Model	#Param	PPL
Shazeer et al. (2014) - Sparse Non-Negative	33B	52.9
Chelba et al. (2013) - RNN-1024 + 9 Gram	20B	51.3
Kuchaiev and Ginsburg (2017) - G-LSTM-2	-	36.0
Dauphin et al. (2016) - GCNN-14 bottleneck	-	31.9
Jozefowicz et al. (2016) - LSTM	1.8B	30.6
Jozefowicz et al. (2016) - LSTM + CNN Input	1.04B	30.0
Shazeer et al. (2017) - Low-Budget MoE	~5B	34.1
Shazeer et al. (2017) - High-Budget MoE	~5B	28.0
Shazeer et al. (2018) - Mesh Tensorflow	4.9B	24.0
Baevski and Auli (2018) - Adaptive Input [◊]	0.46B	24.1
Baevski and Auli (2018) - Adaptive Input [◊]	1.0B	23.7
Ours - Transformer-XL Base	0.46B	23.5
Ours - Transformer-XL Large	0.8B	21.8

Table 4: Comparison with state-of-the-art results on One Billion Word. [◊] indicates contemporary work.

former variants have a large margin over conventional RNN-based models. Notably, our 12-layer architecture achieves the same result as the 64-layer network from Al-Rfou et al. (2018), using only 17% of the parameter budget. In order to see whether better performances can be obtained by increasing the model size, we train 18-layer and 24-layer Transformer-XLs with increased model sizes. With the attention length 784 during training and 3,800 during evaluation, we obtained a new SoTA result and our method is the first to break through 1.0 on widely-studied character-level benchmarks. Different from Al-Rfou et al. (2018), Transformer-XL does not need any auxiliary losses, and thus all benefits are credited to a better architecture.

Similar to but different from enwik8, text8 contains 100M processed Wikipedia characters created by lowering case the text and removing any character other than the 26 letters a through z, and space. Due to the similarity, we simply adapt the best model and the same hyper-parameters on enwik8 to text8 without further tuning. The comparison with previous methods is summarized in Table 3. Again, Transformer-XL achieves the new SoTA result with a clear margin.